

Story

The organisers of PyConES 2026 decided to launch a challenge for all participants: **the Py-rate Radio**. A small speaker connected to an old laptop, with a floppy disk that holds a random message which changes every hour — programmer jokes, talk schedules, hackathon gossip, places to visit in Barcelona, even rumours about the latest NumPy release.

At first it's a fun little game: small shifts, short messages. People laugh as they decrypt them by hand on a piece of paper while waiting for the next talk.

But once the organisers saw how easily the challenges were being solved — and how much the participants were enjoying it — they decided to raise the difficulty: shifts with **six digits**. Yes, six digits. Suddenly, everyone is asking the same question: "*what does this even say?*"

That's when someone opened PyCharm, and the lines of code started running. Yeah.

Problem Statement

Write a **Python** program that reads a string and a non-negative integer k (the shift), and prints the string after applying a **Caesar cipher**: every letter is shifted k positions forward in the alphabet, wrapping around from z back to a (and from Z back to A) as many times as needed.

Uppercase letters must remain uppercase, lowercase letters must remain lowercase. Any character that is not a letter (digits, spaces, punctuation, etc.) must remain unchanged.

Input Format

```
S
k
```

- Line 1: the string S (may contain spaces).
- Line 2: an integer k , the shift.

Constraints

1. $1 \leq \text{len}(S) \leq 500$
2. $0 \leq k \leq 10^6$
3. S may contain uppercase/lowercase letters, digits, spaces, and punctuation.
4. Since k can be much larger than 26, it must effectively be treated as $k \bmod 26$.

Output Format

A single line: the string S after applying the Caesar shift.